Lesson 6: Victory Condition

GameMaker

VIDEO GAME CREATION

This week: a real win condition. Your job is to track how many enemies are left and show "You Win!" the moment the last one is destroyed - completing your game's Level 1 MVP milestone.

Open your own [YourName]Game project from Lesson 5 - don't start a new one. Everything this week builds on the row of enemies you already have.

Tip: no new assets this week - "You Win!" is shown with each engine's own basic text tools, not an image.

Setup

1. Open your [YourName]Game project from Lesson 5.
2. Create a new object obj_gamemanager (no sprite needed) and drag one instance of it into your room, anywhere off to the side - it just needs to exist once, and it's the one place win-tracking will live.

Level 1 - Minimal (MVP)

The moment the last enemy is destroyed, "You Win!" appears on screen. This completes your game's Level 1 MVP milestone - a complete, playable core loop.

3. On obj_gamemanager's Create event, add enemies_remaining = 5; and has_won = false;.
4. Add a Destroy event to obj_enemy - GameMaker automatically runs this once whenever an instance of obj_enemy is destroyed, for any reason. Inside it, decrement obj_gamemanager's enemies_remaining and check for zero (code below).
5. Add a Draw GUI event to obj_gamemanager that draws "You Win!" text when has_won is true.
6. Press F5 (Run) and destroy every enemy to test - "You Win!" should appear the instant the last one is destroyed.

```
// obj_gamemanager's Create event:
enemies_remaining = 5;
has_won = false;

// obj_enemy's Destroy event:
obj_gamemanager.enemies_remaining -= 1;
if (obj_gamemanager.enemies_remaining <= 0) {
    obj_gamemanager.has_won = true;
```

```
}

// obj_gamemanager's Draw GUI event:
if (has_won) {
    draw_text(400, 50, "You Win!");
}
```

Alternate Path: GML Visual (no typing)

The same kill-tracking pattern works with drag-and-drop blocks instead of typed code.

7. On obj_enemy's Destroy event, switch to GML Visual. Drag a "Set Variable (Relative)" block targeting obj_gamemanager's enemies_remaining, then an "If Variable" block checking it against 0, with a "Set Variable" block inside setting has_won to true.
8. On obj_gamemanager's Draw GUI event, drag an "If Variable" block checking has_won, with a "Draw Text" block inside it.
9. Press F5 (Run) and test the same way as the code path.

Level 2/3's per-instance logic already lives the same way in both paths, since GameMaker's Destroy event behaves identically whether it contains blocks or code.

Level 2 - Complete (Alpha/Beta)

The win text appears exactly once, exactly when the last enemy is destroyed - not early, not late, and not repeatedly.

- Destroy all 5 enemies and confirm "You Win!" appears exactly once, exactly when the last one is destroyed - not one enemy early, and not repeatedly.
- This already works correctly because GameMaker's Destroy event fires exactly once per instance, no matter what caused the destruction, and has_won only ever flips from false to true, never back.

Level 3 - Refined (Ready to Publish) - pick any that interest you

- Sound: import the provided sfx_win.wav as a new sound resource (keep its default name, sfx_win), then call audio_play_sound() inside obj_enemy's Destroy event, right where enemies_remaining reaches zero.
- Visual: use draw_set_font() and draw_set_halign() before draw_text() in obj_gamemanager's Draw GUI event to make the win text bigger and centered instead of the small default look.
- Structure: keep all win-tracking (enemies_remaining, has_won) as variables on obj_gamemanager - obj_spawner only ever spawns enemies, and obj_enemy's Destroy event only ever reaches out to modify obj_gamemanager's variables. Neither one draws anything itself.

If Something's Wrong

- "You Win!" never appears: confirm obj_gamemanager actually has one instance placed in the room, that obj_enemy's Destroy event references it correctly, and that it has its own Draw GUI event - a missing Draw GUI event means has_won can be true with nothing ever drawing it.
- "You Win!" appears too early: confirm enemies_remaining's starting value in obj_gamemanager's Create event actually matches your spawn loop's count.

- Error referencing obj_gamemanager: confirm there's actually exactly one obj_gamemanager instance in the room - referencing an object with zero instances by name causes an error.

Want More?

Student Resource List > GameMaker section, or gamemaker.io/en/tutorials